

sg13g2_stdcell_hv — Generation of a Thick-Oxide (3.3 V) Standard Cell Library



IHP SG13G2 open PDK · derivation, all views, verification and physical sign-off

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Scope and result

The IHP SG13G2 open PDK ships no 3.3 V standard cells: its thick-oxide devices appear only in the sg13g2_io pad ring. sg13g2_stdcell_hv fills that gap — **all 84 cells** of sg13g2_stdcell rebuilt on sg13_hv_nmos / sg13_hv_pmos, same topology, pin names and pin order, named sg13g2_hv_* so both libraries coexist in one netlist.

Deliverable	Coverage	Status
SPICE netlist (spice/)	84 cells, 920 devices	verified against 3 independent views

Deliverable	Coverage	Status
CDL netlist (cdl/)	84 cells + 2 tie cells	LVS reference, all 84 cells match
Verilog (verilog/)	84 modules	shared ihp_* UDPs, deliberately not duplicated
xschem symbols / schematics	84 + gallery sheet	netlist-equivalence proven
Qucs-S symbols / schematics	84 + 84 XML + 45 shared .sym	retarget of the thin-oxide views, gated against the SPICE netlist
GDS layout (gds/)	84 cells (66 retargeted + 18 per-cell generated)	DRC clean, LVS clean
LEF abstracts (lef/)	84 macros + CoreSiteHV site	generated from the GDS, pin sets verified against CDL
Liberty NLDM (lib/)	all 84 cells , 668 timing tables, 3 corners	combinational, sequential, tri-state and clock-gate; areas all measured; typ, fast and slow
LibreLane SCL (librelane/)	site, cell maps, tracks, excludes	flops map natively through dfflibmap

Physical sign-off, one fixed invocation of the PDK's own klayout decks — on the abutted two-row array and on the stricter **shared-rail array** (rows at the true 7.14 μm pitch, mixed and mirrored vertical neighbours — what a placed block actually produces) — and, since rev. 5, under Magic's *strict analog* N-well rules as well:

Check	This library	IHP thin-oxide control (same harness)
DRC, cell rules	0	0
DRC, cell rules, shared-rail mixed array	0	—
DRC, Magic (strict analog N-well rules)	0 NW.*	—
DRC, chip-level metal density	7	6
LVS	84 / 84 match	fill cells fail identically without the documented relaxation

The density items are harness properties, not cell defects: metal density is a check on a *filled* die, and the one differing count (M1.j, min 35 % Metal1) reads 35.0 % over the padded array's bounding box against 37.1 % over the actual cell area (control: 45.2 %).

The library has also carried a real block: two variants of the spi_slave IP placed and routed through LibreLane/OpenROAD at 100 MHz pass the full IHP signoff deck — zero geometry violations (section 6).

The device transform

	thin oxide	thick oxide
device model	sg13_lv_nmos / sg13_lv_pmos	sg13_hv_nmos / sg13_hv_pmos
gate length	130 nm	450 nm (Gat . a3 minimum)
NMOS width	—	unchanged
PMOS width	—	× 2.40 , snapped per finger to the 5 nm grid
as/ad/ps/pd	—	recomputed from device geometry
supply	1.2 V	3.3 V

Deliberate long-channel devices (decap 1.0 μm , sighold 700 nm, dlygate4sd3_1 500 nm) keep their lengths; 914 devices moved to 450 nm. The antenna cell is unchanged apart from its name (junction diodes).

Why × 2.40. The thin-oxide library is sized for a centred switching threshold ($V_m/V_{DD} = 0.5046$ at $W_p/W_n = 1.5135$), not drive match. At 3.3 V / 450 nm the PMOS is far weaker (219 vs 533 $\mu\text{A}/\mu\text{m}$) and the same V_m needs $W_p/W_n = 3.63$, i.e. $K_p = 3.63/1.5135 = 2.40$ on every PMOS with every NMOS untouched — preserving each cell’s internal stack ratios. Across the library’s 0.3–17.9 μm width range the required ratio stays within $\pm 5\%$ of 2.40 (work/vm_sweep.py, work/ratio_check.py).

Parasitics follow the vendor deck’s geometry formulas; before any generation they were validated by recomputing all 924 thin-oxide devices (worst relative error 3.75×10^{-4} , the vendor’s four printed digits). The generator refuses to run if this validation fails.

Cost, measured on inv_1 (work/fo4.py): input capacitance 2.67 $\rightarrow$ 5.87 fF (2.20 $\times$), FO4 delay 53.6 $\rightarrow$ 142.4 ps (2.66 $\times$). The thick-oxide NMOS delivers *more* current per micron at 3.3 V than the thin-oxide at 1.2 V (533 vs 391 $\mu\text{A}/\mu\text{m}$) — the loss is capacitance, not drive.

The views

All netlist views are re-emitted from one internal model by work/gen_hv_lib.py, so a device cannot silently diverge between views.

- **spice/** — 84 subckts, 920 PSP103 devices (OSDI/ngspice); widths synchronised to the drawn layout.
- **cdl/** — the same subcircuits as M-cards with *.PININF0, the LVS reference; compared field-by-field against SPICE by verify_sch.py.
- **verilog/** — 84 modules; deliberately no copy of sg13g2_udp.v (the ihp_* UDPs are shared — a second copy would collide).

- **sym/**, **sch/** — two tool front-ends over the same netlist. **xschem**: 84 symbols (thin-oxide drawn geometry, thick-oxide netlist prefix, \$: :SG13G2_HV_SCH resolution) and 84 schematics plus a generated gallery sheet; three-view consistency proven on all 920 devices. **Qucs-S**: 84 schematics, 84 component definitions and the 45 shared gate-shape files, retargeted from the thin-oxide views and gated against the SPICE netlist (see *The Qucs-S views* below).
- **gds/** — **all 84 cells**: 66 by 1-D retarget (section 4), 18 by per-cell generators (section 4.1). Uniform 7.140 μm row height (17 tracks), every width a 0.48 μm site multiple, all contact/via cuts exactly $0.16 \times 0.16 \mu\text{m}$, every rail tap on the site-centred 0.48 μm grid (`work/fix_rail_contacts.py`, section 6).
- **lef/** — 84 macros + CoreSiteHV, generated from the GDS: pins from the pin datatypes labelled by contained text, OBS = drawn metal minus pin geometry, antenna values recomputed from the netlist (the thin-oxide numbers are wrong for 3.5 $\times$ longer gates). Parsed back with klayout and pin sets verified against the CDL.
- **lib/** — Liberty NLDM at three PVT corners: typ 3.30 V/25 $^{\circ}\text{C}$, fast 3.60 V/–40 $^{\circ}\text{C}$ and slow 3.00 V/125 $^{\circ}\text{C}$ (section 7).
- **librelane/** — a complete LibreLane standard-cell-library configuration (section 8), installed as `libs.tech/librelane/sg13g2_stdcell_hv/`.
- **klayout/** — a cell-library registration macro: with the repository's `klayout/` directory on `KLAYOUT_PATH`, all 84 cells appear in KLayout's Instance dialog as library `sg13g2_stdcell_hv`.

The Qucs-S views

The thin-oxide library ships Qucs-S views, so this one does too: 84 schematics in `sch/qucs-s`, and in `sym/qucs-s` one component definition per cell plus the 45 gate-shape files those definitions reference.

They are a device-level retarget rather than fresh drawings. The two libraries are topologically identical — a cell-by-cell comparison against the shipped SPICE netlist confirms it, with exactly two exceptions — so redrawing 920 devices could only introduce differences the transform cannot. Device sizes are read from the netlist, never computed: the thick-oxide transform is not one ratio, because gate length is clamped at 450 nm where the thin-oxide cell used 180 nm, becomes 625 nm where it used 250 nm, and is untouched on the deliberately long devices. The exceptions are `tiehi` and `tielo`, whose thick-oxide topology genuinely differs — a tied-input `inv_1` rather than the thin-oxide four-transistor chain — so they are composed from this library's own `inv_1` with the gate clamped to a rail.

`work/verify_qucs.py` gates the result with three checks, each matched to how the file was produced:

check	scope	criterion
diff against the thin-oxide original	82 retargeted cells	only device size and title-block lines may differ — a changed wire or coordinate fails

check	scope	criterion
device sizes vs the netlist	all 84 cells	920 devices, type and (W, L) per cell
full connectivity	the 2 composed tie cells	netlist extracted from the wire graph, compared terminal by terminal

The first check is a stronger claim than re-deriving connectivity would be: it proves the geometry was never touched, with the thin-oxide library as the authority. Deriving connectivity for all 84 was deliberately not attempted, because 73 of the 920 device instances are rotated or mirrored and a general extractor would have to reimplement Qucs's placement transforms to prove something the diff already proves.

Not verified: that these open and netlist in the Qucs-S GUI. The tool is present in the development container but off PATH, and the PDK's own standard-cell symbol link is broken in that image — see `doc/iic-osic-tools-notes.md`. The files are structurally correct and match the netlist, which is a different and weaker claim.

Layout generation

Hand-drawn 2-D layout cannot be regenerated by placement, but it can be **1-D retargeted**: monotone piecewise-linear coordinate maps applied to every vertex preserve topology, so connectivity survives by construction (`work/layout_retargret.py`). The engineering is in the breakpoints and the exceptions:

- **x-map**: gates widen $0.13 \rightarrow 0.45 \mu\text{m}$ about their centres; contacted poly pitch $0.48 \rightarrow 0.80 \mu\text{m}$. One accepted consequence: `dlygate4sd2_1` staggers two gates at overlapping x, so its PMOS lands at a legal $0.625 \mu\text{m}$, carried consistently into every netlist view.
- **y-map**: PMOS band $\times 2.40$ plus three channel inserts for the thick-oxide clearances (`NW.d1.dig` $0.215 \mu\text{m}$, `pSD.j1` $0.100 \mu\text{m}$, `pSD.i1` rail insert $0.430 \mu\text{m}$), derived library-wide so every cell keeps one row height.
- **PMOS diffusion** is re-emitted per connected slab piece with channel slabs frozen — the only scheme that keeps exact 5 nm-grid device widths without merging, collapsing or renaming devices (each alternative was tried and caught by LVS or `Act.*` rules).
- **Cut layers are translated, never scaled**; contacts are re-tiled inside the new `Activ` at $0.36 \mu\text{m}$ pitch (`Cnt.b1` array spacing), grouped by polygon.
- **ThickGateOx** is drawn on the cell boundary grown $0.27 \mu\text{m}$ in x and $0.42 \mu\text{m}$ in y — the y-margin proven by the N-row edge experiment (the `TGO.a` count stays 2 for any N and moves with the outer edge).
- **Five Metal1 re-routes** (`M1E_EDITS` in the generator) resolve the `M1.e` wide-metal gaps the retarget manufactures; each edit is verified in code (polygon count, `M1.a/b/c1`, 5 nm grid).

- **Netlist ↔ layout sync:** the drawn width is authoritative; `work/sync_netlist_widths.py` brings SPICE and CDL to the drawn geometry per *finger* and recomputes parasitics.
- **Site and tracks:** TRACK_PAD stretches the mid-cell dead zone to make exactly 17 horizontal 0.42 μm tracks; `pad_to_site` pads each width to the 0.48 μm site (13.75 μm total across the library). Mean cell area is **2.87×** the thin-oxide library (median 2.83, range 1.89–4.21).

The 18 cells the library-wide map could not produce

`layout_retarget.py` skips 18 cells, and for a long time they were the library's headline limitation. They are not a single hard case: the skips have **one shared root cause**. A p+ source finger butts into the VDD rail tap, so the library-wide y-map cannot scale that band without moving the rail off the cell boundary and breaking abutment. Several flip-flops add a second obstruction, an NMOS Activ band that reaches the library channel cut.

Once that was understood the fix is mechanical, and is applied per cell rather than library-wide:

1. **Cut the butting finger before the map** — the neck joining the p+ source to the rail tap is removed, asserted exactly, so the band is free to scale.
2. **Split the y-insert per cell** — instead of the library's single 0.975 μm insert, each cell distributes the same total across a rail insert, a channel-cut insert and a pSD insert, chosen so that every template band (rails, taps, pSD, N-well, ThickGateOx) lands exactly on its shipped position and the high NMOS band tops out 0.62 μm below the strict N-well bottom.
3. **Restore the finger after the map** at $y = 6.99 \mu\text{m}$, the butted-junction convention the drawn `slgcp_1` already uses and which the LVS deck connects through `psd_ntap_abutt`.
4. **Rebuild the N-well** with the strict-rule construction of section 6, and re-tile rail contacts onto the site-centred grid.

`work/flop_pilot/gen_seq.py` drives the eight flip-flops from a per-cell insert table; `work/cell_dev/{latches,ebufn,misc}/gen_*.py` cover the four latches, the two tri-state drives, `lgcp_1` and `sighold`. Two cells are not retargets at all: `lgcp_1` is derived from the drawn `slgcp_1` by deleting the scan leg (which also required splitting a shared series node — the scan cell joins the PMOS and NMOS chains in Metal1 where the plain gate does not), and `sighold` is built from scratch on the tie-cell frame. **No device width, length or connectivity deviates from the CDL in any of the 18 cells**, so the existing characterization stays valid; only area changes, and it changes from an estimate to a measurement.

Every generated cell had to pass `work/cell_verify.py` before it was merged into the library: LVS against the CDL, both KLayout decks in an abutted mirrored context, Magic, the structural conventions (row height, site-multiple width, N-well rules) and the pin/track report. The gate was itself validated by running it against an already-shipped cell.

Functional verification

Suite	Coverage	Result
verify_logic.py	60 combinational cells, 452 vectors, both libraries in one deck	PASS
verify_seq.py	16 stateful cells, 400 clocked samples	PASS
verify_sch.py	84 schematics + CDL vs SPICE, 920 devices field-by-field	PASS

The thin-oxide library is the golden reference — the transform changed devices, never logic. The 12 high-impedance states of the tri-state cells are exercised; illegal set/reset combinations are excluded rather than counted as passes; both suites were re-run on the final, shipped netlists.

Physical sign-off

Methodology. Cells are checked in abutted context, never standalone. Two harnesses: `work/make_drc_top.py` (every cell in its own column, mirrored second row at the padded bbox pitch) for cell-internal rules, and `work/make_shared_rail_rows.py` — rows at the true 7.14 μm pitch, orientations N/S/FS, rotated cell order per row, cells advanced by LEF width — for everything only a placed block exhibits. Measurement rules, each bought with a wrong conclusion: one fixed runner invocation (counts from different flag sets are not comparable), diagnosis from flat `klayout.db` rule replication (deep-mode marker coordinates mis-attribute cells), and the IHP thin-oxide library run through the identical harness as control. Pass/fail is parsed from the report database, never the exit code.

DRC. 781 raw violations after the first retarget were driven to zero cell-rule items (density-only residue, same class as the control). One defect class survived the original harness because it could not express it: the retarget left each cell's **rail-tap contacts** at cell-specific x, and in a placed block — where *different* cells share every rail — partially overlapping taps merged into 0.19–0.32 μm bars: ~19 000 Cnt/CntB markers on the first `spi_slave` signoff. `work/fix_rail_contacts.py` re-tiles every rail tap onto the site-centred 0.48 μm grid, which both the FS row flip and placement's x-mirror preserve, with in-code guards for implant polarity, enclosures, gate clearance, contact spacing and tap continuity (1 134 → 1 810 rail contacts).

N-wells: two rule sets, one layout. The retarget placed the N-well for the *digital* rule set — `NW.c1.dig`, 0.31 μm enclosure of HV p-active, which the IHP rulebook grants inside a DigiBnd marker and which the KLayout maximal deck verifies clean. Magic's SG13G2 tech has no DigiBnd concept at all and applies the *analog* rules unconditionally, so it flagged every

cell: 1 128 NW.c1 plus 9 NW.e1 on a placed counter. Measured against the strict rule the library was short by exactly **25 nm in 65 of the 68 cells drawn at that point** (210 nm in mux4_1, 215 nm in slgcp_1, both of which reach further down); the 16 cells drawn afterwards were built to the strict rule from the start.

That margin is recoverable in the well layer alone, so rather than wait on the tool, work/fix_well_nwc1.py rebuilds only the N-well: bottom edge 2.625 → 2.570 μm library-wide, a halo around each cell's PMOS active that forces the two deeper jogs automatically, minus halos around NMOS active and p-taps for NW.d1/NW.f1, and the lateral overhang widened 0.24 → 0.62 μm so the outermost cell of a row is enclosure-clean without a neighbour's merged well (that last point also removed the block-edge NW.e1 flags). Every rule is asserted per cell in the generator with corner-inclusive square sizing, the worst case of the euclidian deck rules.

Nothing about the devices changes. The well is not a device terminal, characterization runs on the schematic netlist, and the HV PSP model cards carry no well-proximity parameters — so no re-characterization was needed, and only the GDS differs. The result: Magic reports **0 NW.*** on the abutted array and both KLayout decks stay clean, i.e. the layout now satisfies the strict and the relaxed interpretation simultaneously. The residual Magic flags (Cnt.c contact-overlap, wide-metal M1.e) are tool-interpretation differences on geometry both KLayout decks accept. The tool gap itself was reported upstream as IHP-Open-PDK issues #1106 (the modular KLayout deck implements neither NW.c1 nor NW.c1.dig, so its “0 errors” was silent on this rule) and #1107 (Magic lacks the DigiBnd relaxation).

LVS. All 84 cells match, re-run in full after the re-tiling. The four device-less fill_* cells need the runner's own --ignore_top_ports_mismatch relaxation — IHP's own fill cells fail the identical flow the same way. LVS caught five defect classes dimensional checks could not (merged channels, collapsed notch devices, mis-paired fingers, a dropped substrate tie, stale diode geometry), which is why the per-cell sweep runs after every layout change.

Block-level validation. The spi_slave IP, in two variants, was synthesised, placed and routed with this library through LibreLane/OpenROAD at 100 MHz on ~354 × 400 μm dies. Both close timing with zero setup/hold violations at the characterised corner, pass routing DRC, antenna and Netgen LVS in the flow, pass gate-level simulation of the routed netlist (all four SPI modes on the all-modes variant), and pass the **full IHP signoff deck** with zero geometry violations — only the 8 chip-level density/filler markers a filled die resolves at tapeout. The flow settings this takes (flip-flop mapping onto sdfbbp_1, excluded cells, Metal1 routing, rails-only fillers) are documented in the spi_slave-ihp repository.

Liberty characterisation

lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib, lib/sg13g2_stdcell_hv_fast_3p60V_m40C.lib and lib/sg13g2_stdcell_hv_slow_3p00V_125C.lib are produced with **CharLib** against the shipped,

layout-synchronised SPICE netlist on the PSP103/OSDI models — 25 839 combinational simulations plus a 2 962-task sequential run: 600 delay/slew tables over 66 cells (52 combinational, 9 flip-flops, 5 latches), none empty. The 6 tri-states add 60 more and the 2 statetable clock gates 8 more, both from the direct measurements of section 7.2, for 668 tables over 74 cells with arcs in each shipped corner. Grids are the thin-oxide grids rescaled by the measured $2.66\times$ delay / $2.20\times$ capacitance ratios; boolean functions are translated from the thin-oxide Liberty and checked by truth-table equivalence. Stock CharLib needed five committed tool workarounds (OSDI shim, ngspice-shared backend, case-insensitive supply lookup, Liberty syntax post-pass) and — for sequential cells, which stock CharLib cannot characterise at all — custom $\text{clk}\rightarrow\text{Q}$ and setup/hold-bisection procedures with a c2q-degradation pass criterion (`work/seq_delay_procedure.py`).

The shipped file is verified **as data** by `work/verify_lib.py`: structure (668/668 tables populated at each of the three corners), cross-view (all pins exist in the CDL), physical (every area equals the drawn boundary to 1 nm^2), monotonicity along the load axis (2 169/2 170 series, one documented pessimistic waiver at typ — `xnor2_1` glitch latching — with fast and slow clean), input capacitance against an independent both-rails measurement (6.44 vs 5.87 fF, 9.7 % — this check caught a $6.8\text{--}7.5\times$ Miller defect in CharLib's default procedure), and sequential arcs (14/14 cells: delay + setup + hold present, nothing pinned at search bounds). A hand-written transient at a table corner agrees to 0.3 %.

An independent characteriser, **lctime**, was run over eight cells on the same grids and models: in the region STA exercises (real loads, slews $< 1\text{ ns}$) the two agree to a median 2.9 % on delays and 0.0 % on transitions; every disagreement region traces to a stimulus convention, and a direct measurement sides with the shipped table (+0.15 % vs -19% at the probe point). lctime's input capacitance (+52 %) also corroborates validating C_{in} against a direct measurement, not a characteriser.

`tiehi/tielo` are present in the Liberty without timing arcs (a constant output has none) and with directly measured leakage — `work/tie_leakage.py` applies the sequential cells' settled-tail average to the tie netlists ($0.011 / 0.013\text{ nW}$), so no number is borrowed from `inv_1`.

Drive limits, and why their absence is not benign

CharLib emits no `max_capacitance` or `max_transition` at all, and the first three revisions of this library shipped without them. That is worse than a missing convenience. OpenSTA answers “no limit” for every pin, so a flow's max-cap and max-slew checks pass **vacuously** — reporting zero violations because there is nothing to violate — and OpenROAD's TritonCTS dereferences an empty buffer-selection result and **segfaults** during clock-tree synthesis, which is how the gap surfaced.

`work/finalize_lib.py` derives the limits from the characterization itself rather than copying the thin-oxide numbers: a pin may not be asked to drive more load than its tables cover, nor to accept a slower edge than was characterized. Per output pin `max_capacitance` is the top of its load axis ($0.66\text{--}10.56\text{ pF}$ by drive class), per pin `max_transition` the top of the slew axis it was characterized against (6.67 ns combinational, 3.36 ns sequential), with library defaults

following the thin-oxide convention of weakest-drive / slowest-edge. The crash is reproducible on demand and was reported upstream with a minimal two-flop test case (OpenROAD issue #11165): a liberty without limits should produce a diagnosable error, not a signal 11.

The cells CharLib cannot reach

Nine cells are drawn but were never characterized, and the reason is a tool limitation in each case rather than a property of the cell: `gen_charlib_config.py` skips the two statetable clock gates (CharLib has no statetable input form) and the device-less/no-output cells, while the six tri-states are *configured* but silently produce nothing — CharLib has no high-impedance concept anywhere in its schema or code, so it emits an empty result without an error. `lctime` is no substitute: it can carry a `three_state` attribute but deliberately pins the enable and skips exactly the enable arcs one needs.

They are therefore measured directly against the shipped netlists, the same way the tie and sequential cells already are.

sighold (bus holder) is the simplest and is complete: it has no timing arcs at all — the thin-oxide model is `driver_type : bus_hold` plus capacitance and per-state leakage — so it needs leakage (0.0118 nW / 0.0224 nW, $\sim 10^4$ below the thin-oxide cell, the same thick-oxide ratio the tie cells show) and a pin capacitance. The capacitance is the interesting part: charge integration, the library's method for ordinary gate pins, is **invalid here**. On a bus holder the integral is dominated by the keeper fighting the driver, so it grows monotonically with the driving edge rate — measured 30.5 fC at a 0.2 ns edge to 86.4 fC at 5 ns, a 2.8× spread — and is a property of the driver, not the cell. A 1 MHz small-signal measurement near the rails gives 0.00367 pF with a 7 % spread across bias; the keeper fight-back charge is recorded separately rather than folded in. The same reasoning explains the thin-oxide cell's 2.8× rise/fall capacitance asymmetry as an artifact of its measurement, so the HV cell ships equal rise and fall values.

The six tri-states are characterized by `work/char_tristate/char_tristate.py` and are in the shipped Liberty. Each carries three arc classes rather than one: the combinational data arc, plus `three_state_enable` and `three_state_disable` measured into and out of a floating node defined by a 1 GΩ mid-rail keeper — the construct the functional suite already uses for its 12 Hi-Z checks. The harness is calibrated by re-measuring a shipped cell's pin capacitance through the same code path (0.46 % against the shipped value), and the combinational arcs land inside the library's 2.66× delay band.

The disable arc required a change of measured quantity, not just of threshold. Waiting for a released output to cross a voltage threshold is not physically measurable: once the driver lets go, only the keeper moves the node, so the answer is $0.2 RC \approx 0.5$ ms and scales with the load. The thin-oxide library cannot have measured it that way either — its disable tables are exactly load-independent, differing by a 1 fs monotonicity epsilon. What is measured here is the drive current going away: from `TE_B` at 50 % to $|I_z|$ falling through half its on-state value, which is load-independent by construction and needs no synthetic epsilon. Section 10.2 of the companion characterization report gives the derivation and the ratio tables.

The two clock gates are measured by `work/char_clockgate.py`: a CLK→GCLK propagation arc, `setup_rising/hold_rising` on the enable pins — the form the sequential bisection procedure already emits — and a `min_pulse_width` constraint on the clock pin, bisected until the gated output loses its pulse, alongside the `statetable`, internal state pin and `state_function` metadata that make an integrated clock gate usable. With these the Liberty covers **all 84 cells**.

The min-pulse-width measurement is worth one caution, because its first result was wrong in a way that looked right. Under machine load the trials time out; a timed-out trial returns no measurement, the bisection predicate reads that as “failed at every width”, and the search then returns its own upper bound. The first fragment carried a 48 ns minimum pulse width — 50× the cell’s own CLK→GCLK delay — which would have told STA that every realistic clock pulse is too narrow. The generator already recorded a bracketed flag for exactly this case but emitted the value anyway; it now refuses to, and names the offending points. All 16 points are bracketed in the shipped data, 0.46–1.11 ns and monotone in input slew.

`lgcp_1` and `slgcp_1` come out with identical min-pulse-widths, which is expected rather than suspicious: their CLK→GCLK delays differ only in the 4th–6th digit and their clock-pin capacitances in the 7th — the scan leg sits in the enable path, not the clock path — so the true difference is far below the 0.01 ns bisection tolerance and both converge to the same midpoint. That the two simulations are genuinely distinct is visible in the delay and capacitance data.

Both cells nevertheless stay in the exclude lists, as the thin-oxide SCL keeps its own: an integrated clock gate is instantiated deliberately, not inferred.

One deliberate omission: switching power

This library ships **no internal_power tables for any cell** — the characterization was scoped to timing and leakage. The special-cell work above kept that scope rather than adding power for nine cells alone: a library where only the tri-states and clock gates carry switching power, and every other cell reports zero, produces power analysis that is worse than uniformly absent because it looks complete. It is recorded as a library-level limitation instead.

LibreLane integration

A library that a digital flow cannot load is not usable, and until rev. 5 this one could not be loaded: LibreLane rejects any `STD_CELL_LIBRARY` without a `libs.tech/libreLane/<name>/` directory, and the PDK-level `config.tcl` hardcodes the thin-oxide liberty **filenames** in its `LIB` dict while validating every path eagerly. The library now ships the missing pieces.

The SCL (`libreLane/`, installed as `libs.tech/libreLane/sg13g2_stdcell_hv/`): `config.tcl` with the CoreSiteHV site (0.48×7.14) and the HV driving, tie, fill, decap, diode and CTS cells; `tracks.info`; latch, mux and tri-state techmaps; and both exclude lists, which mirror the thin-

oxide ones. PDN_RAIL_WIDTH stays 0.44 μm — the VSS rail is 0.44 μm symmetric about the row edge and the VDD pin shape, though taller, fully covers a 0.44 μm strap.

The PDK config gets a conditional block for STD_CELL_LIBRARY == sg13g2_stdcell_hv (every characterized corner is registered; *_typ_3p30V_25C is the default, VDD_PIN_VOLTAGE 3.30), and the library ships a copy of the shared sg13g2_tech.lef because that config globs it per-SCL and a Tcl glob errors rather than returning empty. Both patches are idempotent.

Flip-flops now map natively. Before all 84 cells were drawn, the only flop with both a liberty and a layout view was the scan cell sdfbbp_1 — which dfflibmap cannot target, because its next_state, (SCE & SCD) | (!SCE & D), is not a plain D function. The interim solution was sdfbbp_map.v, a techmap covering every posedge Yosys flip-flop type on the scan cell, with the scan mux recycled as the enable mux for \$ _DFFE_* and sync resets folded into the D leg; all 23 clocked mappings were proven equivalent to the Yosys cell semantics with equiv_induct. With the dfrbp/dfrbpq/sdfrbp/sdfrbpq layouts drawn it is no longer needed — dfflibmap maps directly, as in the thin-oxide flow — and it remains only as a documented opt-in for DFT flows that want every flop on the scan cell: drop sg13g2_hv_sdfbbp_1 from synth_exclude.cells and set, at design level, SYNTH_EXTRA_MAPPING_FILE: pdk_dir::libs.tech/librelane/sg13g2_stdcell_hv/sdfbbp_map.v (LibreLane does not let a PDK set that variable). Removing it cut sequential area on the reference counter by **31 %** (1 699.89 $\rightarrow$ 1 178.96 μm^2), the tie-off overhead the workaround carried.

One lesson worth recording, because it cost two debugging rounds: a techmap applied *after* the synthesis pass's own lowering must write its glue as fine-grained gates (\$ _NOT_, \$ _AND_, \$ _OR_). Inline expressions such as ~E create coarse cells at a point where no later pass will lower them, and they reach the netlist unmapped. The thin-oxide latch map has the same latent issue; both HV maps avoid it.

A view can be present and still be missing. The spice view was absent from the first three revisions of the upstream PR even though the generator wrote it and the install script copied it every time: the PDK repo's root .gitignore carries *.spice, and its libs.ref/.gitignore re-includes only the spice/ directory, not the file inside it, so git add skipped it silently. The thin-oxide library's own spice view predates that rule and is already tracked, so nothing complained — and the failure reproduced only for someone cloning the branch, never for anyone testing an installed tree, which is why every local gate passed. make_pdk_pr.py now asks git check-ignore --no-index about every installed file and fails the run if any would be invisible; the check reproduces the defect against the unpatched .gitignore and passes 410 files against the fixed one.

Tri-states are wired too, now that they carry timing: SYNTH_TRISTATE_MAP maps Yosys' \$ _TBUF_ onto sg13g2_hv_ebufn_2 and TRISTATE_CELLS lists both footprints, mirroring the thin-oxide SCL.

Block validation. The reference 8-bit counter from the IIC-JKU SG13G2 AMS chip template hardens end to end on the shipped library with no design-level liberty or corner overrides: flops mapped natively, zero unmapped cells, CTS clean, max-cap/max-slew checks non-

vacuous and passing, LVS clean through the auto-derived CELL_SPICE_MODELS, and **both** KLayout and Magic DRC at zero.

A second design exercises the tri-states specifically — two registered sources driving one shared internal bus, so synthesis must infer `$_TBUF_` and STA must time the enable arcs. It hardens equally clean: 16 `sg13g2_hv_ebufn_2` instances, zero unmapped cells, Magic DRC 0, KLayout DRC 0, LVS 0, routing DRC 0, antenna 0, and zero max-slew and max-cap violations against real limits.

Verification summary

Check	Method	Scope	Result
Parasitic formulas	recompute vendor as/ad/ps/pd	all 924 thin-oxide devices	PASS, worst error 3.75×10^{-4}
Combinational logic	ngspice vs thin-oxide golden	60 cells, 452 vectors, 12 high-Z states	PASS
Sequential logic	ngspice, clocked walk from reset	16 stateful cells, 400 samples	PASS
Three-view consistency	symbols vs SPICE vs CDL	84 cells, 920 devices	PASS
DRC, padded array	PDK deck, fixed invocation	84 cells	0 cell rules ; 7 density items
DRC, shared-rail array	same deck, true-pitch mixed/mirrored rows	84 cells × 4 rows	0 cell rules ; density only
DRC control	identical harness on IHP <code>sg13g2_stdcell</code>	84 cells	0 cell rules, 6 density
DRC, Magic strict analog rules	<code>drc(full)</code> , euclidean, abutted array	84 cells	0 NW.* ; Cnt.c/M1.e tool-interpretation only
LVS	PDK deck, per cell, after the well rebuild	84 cells vs CDL	84/84
LEF	klayout parse-back + pin sets vs CDL	84 macros	PASS, on-grid
P&R block validation	LibreLane <code>spi_slave</code> ×2 + full signoff deck	~354 × 400 μm, 100 MHz	all clean; signoff 0 geometry violations , 8 density markers
Liberty structure/views/areas	<code>verify_lib.py</code> 1–3	84 cells , 668 tables	PASS, all areas measured , exact to 1 nm ²
Liberty monotonicity	<code>verify_lib.py</code> 4	2 170 delay series	2 169 + 1 documented waiver (typ only; fast and slow are clean)

Check	Method	Scope	Result
Liberty Cin	vs both-rails reference	inv_1	6.44 vs 5.87 fF (9.7 %)
Liberty delay point-check	hand-written transient	table corner	0.3 %
Independent characteriser	lctime, 8 cells, 3 132 points	STA region	median 2.9 % / 0.0 %
Sequential arcs	verify_lib.py 6	14 flip-flops/latches	14/14 clean
Liberty, all corners	verify_lib.py per corner	typ, fast, slow	3/3 PASS , 84 cells and 668 tables each
Qucs-S views	verify_qucs.py diff + sizes + connectivity	84 schematics, 920 devices	PASS
Site geometry	boundary scan	all 84 cells	7.140 μm , widths on 0.48 μm site
Tie-cell leakage	tie_leakage.py settled-tail average	tiehi/tiel0	0.011 / 0.013 nW, in the Liberty
Pin track alignment	grid_align_pins.py --apply + full re-signoff	12 of 25 off-track pins widened; 11 of 282 remain	DRC 0 cell rules $\times 2$, LVS clean
PDK dev-branch re-run	IHP-Open-PDK dev decks (KLayout 0.30.9)	both DRC arrays, 84-cell LVS	main tables clean , LVS 84/84
New-cell signoff gate	cell_verify.py per candidate before merge	18 generated cells	18/18 PASS (LVS + both decks + Magic + structure + pins)
Liberty drive limits	verify_lib.py 7	79 output pins	present on every pin; TritonCTS crash resolved
Liberty special classes	verify_lib.py 8	tri-state / ICG / bus-hold constructs	complete-construct check
LibreLane block run	reference counter, no design-level overrides	RTL→GDS	flops native, DRC 0/0 , LVS clean
LibreLane tri-state run	shared-bus design inferring \$ _TBUF_	RTL→GDS	16 ebufn_2, DRC 0/0 , LVS 0, slew/cap 0
Git visibility of views	git check-ignore --no-index in make_pdk_pr.py	410 installed files	all trackable

The library is submitted upstream as [IHP-Open-PDK PR #1103](#) (draft): `libs.ref/sg13g2_stdcell_hv` following the dev-branch layout, a two-line xschemrc registration, and an optional KLayout autorun macro, with the cell-set choice, the macro, and xschem view-

machinery unification flagged as maintainer questions. `work/make_pdk_pr.py` assembles and verifies the contribution against a checkout.

Known limitations

- **No switching power.** The library ships `leakage_power` but no `internal_power` for any cell (section 7.3). Dynamic power analysis will see only leakage.
 - **Clock gating is available but not automatic.** `lgcp_1` and `slgcp_1` carry the full ICG model, but an integrated clock gate is instantiated deliberately rather than inferred, so both remain in the exclude lists exactly as the thin-oxide SCL keeps its own. Enabling it needs two design-level variables, documented in the exclude list.
 - **Three corners** (typ 3.3 V/25 °C, fast 3.6 V/−40 °C, slow 3.0 V/125 °C); sequential leakage is a single-settled-state number; the delay cells' ratios shift with the 450 nm minimum; the decaps store less per unit area.
 - **Timing is schematic-characterized.** Device sizes match the drawn layout exactly (LVS-proven per cell), but no parasitic extraction is folded back into the tables; a PEX-based re-characterization is the natural next step.
 - **11 signal pins remain off the vertical Metal1 track grid** and cannot be widened within `M1.b` (12 others were). Route with `RT_MIN_LAYER Metal1`, or expect the router to jog.
 - **Not silicon-proven.** Everything here is simulation against the PDK models. The library has now been independently re-tested by a third party — RTL-to-GDS on the upstream PR, reproducing the DRC, LVS, STA and mapping results — but it has not been fabricated.
-

Reproducing the library

```
cd /foss/designs/sg13g2_stdcell_hv/work

python3 gen_hv_lib.py          # netlists, schematics, symbols, Verilog
python3 gen_gallery.py         # gallery sheet
python3 layout_retarg.py       # gds/ (66 cells; prints the 18 skips)
python3 gen_tie_cells.py       # + tiehi / tielo
python3 flop_pilot/gen_seq.py   # the 8 flip-flops (per-cell y-map recipe)
# latches, ebufn drives, lgcp_1, sighold: cell_dev/*/gen_*.py
python3 cell_verify.py <cell>.gds # gate every generated cell before merge
python3 fix_rail_contacts.py    # rail taps onto the site-centred grid
python3 sync_netlist_widths.py  # SPICE + CDL follow the drawn geometry
python3 fix_well_nwcl.py        # N-wells to the strict analog HV rules
python3 gen_lef.py             # lef/ (CoreSiteHV + 84 macros)
python3 make_drc_top.py         # padded two-row DRC context
python3 make_shared_rail_rows.py # shared-rail mixed-row DRC context

python3 verify_logic.py; python3 verify_seq.py; python3 verify_sch.py
./run_lvs.sh                   # per-cell LVS, 84/84
# DRC: PDK run_drc.py on drc/drc_top.gds and drc/shared_rail.gds
#      plus magic drc(full) on drc_top.gds for the strict N-well rules

# Characterization is per PVT corner (corners.py); one command per corner:
./run_corner.sh typ            # or fast | slow
# CharLib for what it can express, the project's own procedures for the
# sequential cells, direct ngspice measurement for the cells no
# characteriser models (tri-states, clock gates, bus holder, tie cells),
# then drive limits, areas and the data gate -- in that order, because
# the drive limits are derived from the table axes of whatever is
# already merged.
python3 lctime_compare.py      # independent characteriser cross-check

python3 make_pdk_pr.py --pdk <IHP-Open-PDK checkout> # assemble the PR
```

Every number in this report is reproducible from these scripts; the `work/` directory and the `README.md` are the provenance of record.